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Coupling scheme and probe damper for pulsed nuclear magnetic resonance single coil probe*

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A coupling scheme and a probe damper for use with a pulsed nuclear magnetic resonance single coil probe are described. The designs for both circuits incorporate the unique properties of PIN diodes. The performances of the circuits at the signal processing level are evaluated at 4.3, 15.78, and 49.0 MHz.

I. INTRODUCTION

A fundamental problem associated with the design of pulsed nuclear magnetic resonance (NMR) spectrometers is the coupling of the transmitter, probe, and receiver. During the time interval that large radio frequency (rf) pulses are applied to the probe, the sensitive receiver must be protected to avoid saturation and/or damage. Within microseconds after the rf pulses, the receiver must be capable of amplifying small nuclear resonance signals from the probe, preferably with the transmitter isolated. Ideally the coupling scheme should perform both functions with small rf losses.

In the course of the development of a high sensitivity pulsed magnetic resonance spectrometer to study gas phase nuclear relaxation, we investigated the coupling schemes reported in the literature and found none of them to be adequate. The use of crossed coils has been discussed by Mansfield and Powles,¹ Clark,² and Jones *et al.*³ Typical isolation achieved is 60 dB. The probe design problems associated with the use of separate and orthogonal transmitter and receiver coils are substantial. Bridge techniques have been developed by Lowe and Barnaal,⁴ Jeffrey and Armstrong,⁵ Klein and Phelps⁶ and Moores and Armstrong.⁷ Unfortunately bridge techniques suffer either from rather critical tuning requirements or from insufficient isolation, and from inherent 6 dB losses both in transmitter power and in nuclear signal coupled to the receiver. Loose coupling schemes have been reported by Trappeniers *et al.*,⁸ Gray *et al.*,⁹ and McKay and Woessner.¹⁰ Although simple in design, these schemes provide inadequate matching and isolation capabilities. A direct coupling network was devised by Lowe and Tarr¹¹ and extended to the vhf range by Moores and Armstrong.⁷ Although a relatively effective scheme, the isolation of the transmitter and receiver were not sufficient.

In Sec. II we describe a direct coupling scheme which employs transmission line properties and two PIN diode gates, one for the transmitter and one for the receiver. The principles in this coupling network represent an extension of the ideas of Lowe and Tarr.¹¹ Its operation is evaluated at 4.3, 15.78, and 49 MHz.

The time constant τ associated with a resonant LCR circuit is given by

$$\tau = Q/\pi\nu_0,$$

where ν_0 is the resonance frequency. Following a transmitter

pulse of $\sim 100 V_{\text{rms}}$ into the resonant circuit, about 20 time constants are required for the ringing to decay to the level of a small but measurable nuclear signal of say $0.2 \mu V_{\text{rms}}$. For experiments in which sensitivity requirements call for a high Q circuit and where $\nu_0 \lesssim 50$ MHz, the receiver dead time due to coil ringing following a transmitter pulse can severely reduce the amount of nuclear signal available for processing. The shorter the nuclear free induction decay time constant, the greater the reduction of the signal before it can be observed.

In such cases, there is need for a probe damper, a device which will allow normal operation of the resonant circuit at all times except for short periods following the transmitter pulses during which the energy in the resonant circuit is quickly dissipated. A number of such devices have appeared in the literature. Clark,² Jones *et al.*,³ and Samuelson and Ailion¹² describe devices for use with crossed coil systems. Spokas¹³ reported on a device for use with either crossed coils or a bridge technique. Lowe and Barnaal⁴ have suggested a complex bridge circuit for use with single coil systems. Trappeniers *et al.*,⁸ McKay and Woessner,¹⁰ and Roeder *et al.*,¹⁴ have presented devices for use with single coil circuits. For none of these circuits was the damping factor exceptional. Furthermore, performance evaluations at the signal processing level for the measurement of small signals are lacking. Such evaluations are essential to show low level spurious effects.

In Sec. III we describe a probe damper for use with a single coil circuit based on the application of a PIN diode as a current controller resistor. The performance of the circuit for the observation of small signals is evaluated.

Section IV summarizes the pulse requirements needed to activate the coupling scheme and the probe damper.

The presentation concentrates on the more novel aspects of the work. For additional details the reader is referred to the Ph.D. thesis of Kisman.¹⁵

II. COUPLING SCHEME

A block diagram of the coupling scheme is shown in Fig. 1. The principle of operation is simple. During the transmit mode, the transmitter gate conducts the rf pulses directly to the probe while the receiver gate is closed. There is little loss of power and the receiver is well protected. During the receiver mode, the transmitter gate is closed and the

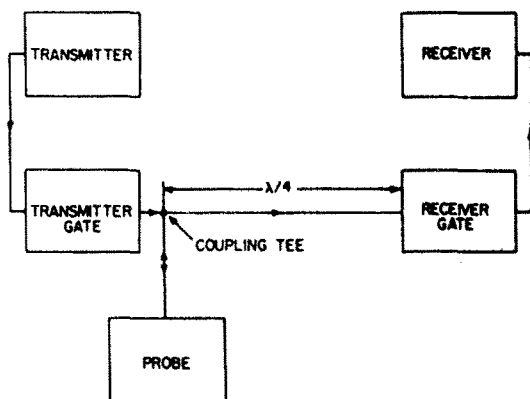


FIG. 1. Block diagram of the coupling scheme linking the transmitter, probe, and receiver. The rf length of the transmission line before the receiver switch must be an odd multiple of $\lambda/4$, where λ is the nuclear Larmor wavelength.

receiver gate opens to allow the nuclear signal to pass to the receiver. The transmitter noise is isolated and little nuclear signal is lost.

The coupling scheme has a frequency range of <3 to >100 MHz. The only frequency dependent components are four coaxial cables, the one indicated in Fig. 1, and three others within the gates. They have a bandwidth of $>15\%$ for a decrease in the isolation of the gates of 6 dB.

Impedance matching is straightforward. It is only necessary to match the transmitter, the receiver, and the probe individually to the cable impedance. The insertion of the coupling scheme requires no additional adjustments.

The transmitter gate is shown in Fig. 2. Unitrode 4001 PIN diodes are used. The configuration is possible through the use of ferrite bead chokes which isolate the driver from the rf. These chokes provide a short to dc but a large and mainly resistive impedance to rf.

In the transmit mode all the PIN diodes are biased on with -120 mA and present only $\sim 0.2 \Omega$ impedance to rf. The small impedance Z_{PIN} of diode C is transformed to a large impedance Z_T at point T by the $\lambda/4$ transmission line, viz.

$$Z_T = Z_0^2 Z_{PIN}^{-1},$$

where Z_0 is the characteristic impedance of the transmission line. As a result, the transmitter pulses pass through the

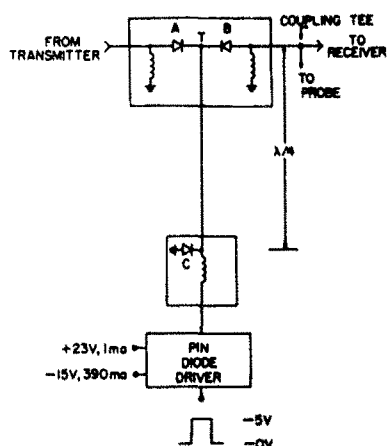


FIG. 2. The transmitter gate within the coupling scheme.

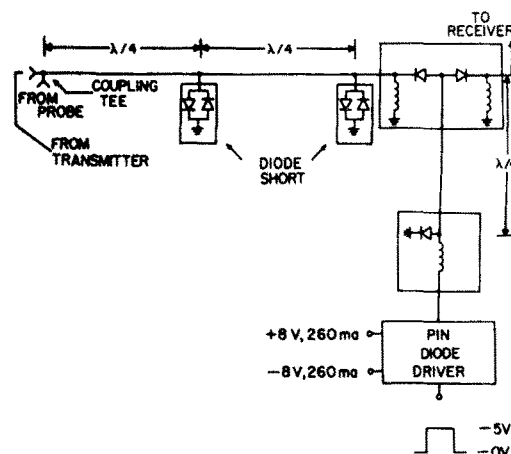


FIG. 3. The receiver gate within the coupling scheme.

gate with little attenuation. Note that the PIN diodes are arranged to avoid back biasing against the large transmitter pulses. This arrangement provides high rf power capabilities with relatively small dc bias.

In the receive mode, all the diodes become back biased by $+23$ V and present large impedances to rf (determined mainly by the capacitive impedance Z_c of the diodes). Now diode C reflects a small impedance at T. In this mode the gate isolates the transmitter from the probe and the receiver. The isolation in dB may be approximated by¹⁶

$$A \approx 10 \log \left(\frac{Z_c}{Z_0} \right)^2 + 10 \log \left(\frac{Z_c}{2Z_T} \right)^2, \quad (1)$$

where Z_0 is the input and output impedance and Z_T is the impedance at T. Using 4000 series diodes, the isolation predicted varies from 130 dB at 4 MHz to 79 dB at 49 MHz. This result applies until the peak rf voltage becomes larger than the back bias voltage; thereafter the isolation should decrease relatively slowly.

The receiver gate is depicted in Fig. 3. The PIN diode configuration is identical to that in the transmitter gate except the diodes used were Unitrode 7201 and their polar-

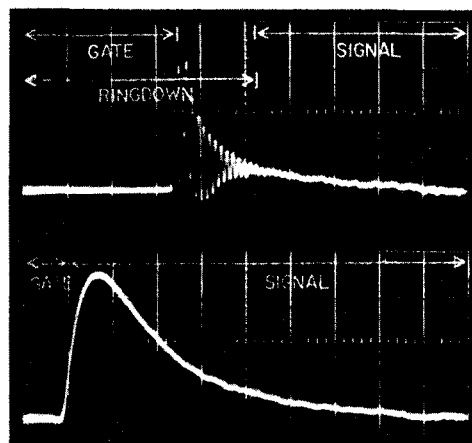
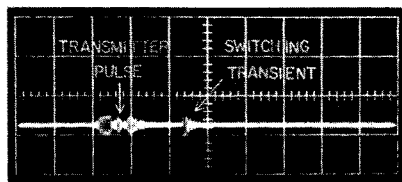


FIG. 4. A ^{19}F Teflon signal at 4.3 MHz. The horizontal scale is $20 \mu\text{sec/div}$. (a) With no transmitter gate, ringing from the transmitter tank circuit swamps the nuclear signal for $>100 \mu\text{sec}$; (b) with the transmitter gate, no evidence of the transients remains; the nuclear signal is clearly visible.

FIG. 5. The output of the preamplifier in the receiver as observed at 15.78 MHz. The vertical scale is 0.05 V/div; the horizontal scale is 5 μ sec/div.



ities are reversed. The diode shorts to ground consist of several crossed pairs of 1N4151 silicon diodes.

In the transmit mode, the large rf pulses at the coupling tee see a large impedance in the receiver branch. This results from the transformation by the $\lambda/4$ transmission line of the small impedance presented to large rf voltages by the first diode short to a large impedance at the coupling tee. If one diode short does not provide sufficient isolation, another can be added (as shown in Fig. 3). Once the rf has been reduced by diode shorts to the level of the reverse bias voltage applied to the PIN diode switch, this switch provides substantial additional isolation.

In the receive mode, the small nuclear signal is conducted to the receiver with negligible loss. The signal is too small to bias on the diode shorts so that the only loss through these shorts results from the capacitive impedance of the diodes (~ 2 pF per diode). This restricts the number of diodes one can use (e.g., at 49 MHz, 8 diodes result in a 1 dB loss of signal). In the PIN diode switch, the diodes are biased so that they appear as small resistances to the signal. They cause little loss of signal and contribute little to the noise.

Figure 4 illustrates the effectiveness of the transmitter gate. The photographs show a ^{19}F teflon signal at 4.3 MHz as obtained from the receiver. Without the transmitter gate in operation [Fig. 4(a)] ringing from the transmitter, tank circuit overwhelms the nuclear signal for >100 μ sec. With the transmitter gate in operation [Fig. 4(b)], no trace of the transients from the transmitter remain and the nuclear signal is clearly visible. The long risetime is expected for a nuclear signal induced in a high Q resonant circuit. The signal begins to grow only after the removal of the probe damper pulse.

The transmitter gate provides isolation, from a 3 V_{rms} input signal, of 76 dB at 49 MHz, 98 dB at 15.78 MHz, and

110 dB at 4.3 MHz. A discrepancy with the performance predicted by Eq. (1) results from stray effects, once the isolation becomes greater than 100 dB. The attenuation of the transmitter pulses passing through the gate varies from 0.5 to 1.4 dB. The PIN diode switch within the receiver gate provides isolation from a 3 V_{rms} input signal of 69 dB at 49 MHz, at 15.78 MHz, and 109 dB at 4.3 MHz. The attenuation to nuclear signals varies from 0.6 to 0.9 dB.

The performance of the entire receiver gate under experimental conditions is illustrated in Fig. 5. The photograph shows the output of the preamplifier in the receiver at 15.78 MHz. The remnants of both a 300 V_{pp} transmitter pulse and a transient switching signal are observed to be less than 0.04 V_{pp} so that the isolation at the coupling is 98 dB. The isolation observed at 4.3 and 49 MHz is 142 dB and 105 dB, respectively. With this receiver gate, no saturation of the first preamplifier in the receiver occurs at any frequency.

III. PROBE DAMPER

A block diagram of the probe damper circuitry is given in Fig. 6. In essence, the damper consists of a PIN diode (Unitrode, 4001) placed between the tuning capacitor and ground together with the associated electronics required for active control of the diode. The parallel resonant circuit used contained split capacitor matching, with the ground and 50 Ω points reversed from that normally employed, for ease of probe construction. Tuning is mainly achieved by means of the coaxial cylindrical capacitor C_2 and matching via capacitor C_T below the Dewar. The coil and tuning capacitor are contained in a copper cylinder which serves as an electrical shield. C_1 and C_3 are the distributed capacitances across the coil and between the tuning capacitor and ground, respectively. The PIN diode must be out of the magnetic field although it can operate over a very wide temperature range. With the PIN diode in place, the resonant circuit is well represented by the circuit in Fig. 7(a). In normal operation, R_p is effectively the small forward bias resistance of the PIN diode and the C_1 , C_2 , C_3 , R_p network reduces to C_1 and C_2 in parallel. During probe damping, R_p is the large reverse bias resistance of the PIN diode in parallel with R_R . The C_1 , C_2 , C_3 , R_p network can be approximated by a simple

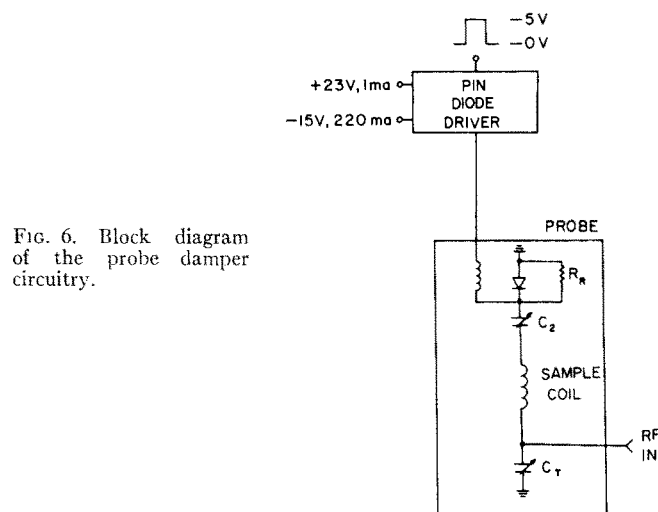


FIG. 6. Block diagram of the probe damper circuitry.

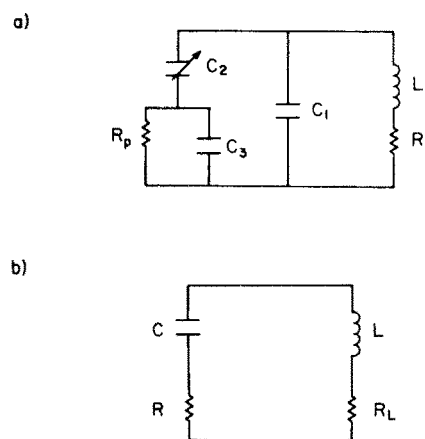


FIG. 7(a). The resonant circuit in the probe; (b). equivalent circuit for the resonant circuit.

series RC combination [Fig. 7(b)]. The impedance matching capacitor may be neglected in the analysis. The problem is to optimize the C_1 , C_2 , C_3 , R_p network so as to maximize R and therefore to minimize the ringdown time constant,

$$\tau_{\min} = \frac{2L}{(R_{\max} + R_L)}$$

It follows that

$$R_{\max} = \frac{1}{2(C_1/C_2 + 1)^2} (R_p)_{\text{opt}},$$

where

$$(R_p)_{\text{opt}} = \frac{C_1 + C_2}{\omega(C_1 C_2 + C_1 C_3 + C_2 C_3)}.$$

The reduced ringdown time constant τ_{PD} is given by

$$\tau_{PD} \approx \left(\frac{R_L}{R_{\max}} \right) \tau_0,$$

where τ_0 is the ringdown time constant in the absence of a probe damper. Note that probe damping is enhanced by minimizing C_1 and C_3 and by setting $R_p = (R_p)_{\text{opt}}$.

The best performance of the probe damper is expected at 4.3 MHz. At this frequency $Q \sim 80$ and for $R_p = 1.1 \text{ k}\Omega$, the decay time constant is expected to be reduced by a factor of more than 300. The observed performance is shown in Fig. 8. A 1 atmosphere hydrogen gas signal filtered using a $1 \mu\text{sec}$ filter and signal averaged 1024 times is observed. The application of the probe damper has reduced the dead time from $>140 \mu\text{sec}$ [Fig. 8(a)] to $\sim 18 \mu\text{sec}$ [Fig. 8(b)]. This represents a damping factor of ~ 8 . Note that the long risetime is expected for a nuclear signal induced in a high Q resonant circuit. The theoretical reduction factor at this frequency is ~ 300 ; it is not achieved for at least two reasons. First, maximum effectiveness is expected only after the ringdown has reached $\sim 25 \text{ V}$. Second, the fall time of the driving pulse must be slow enough to avoid serious transient

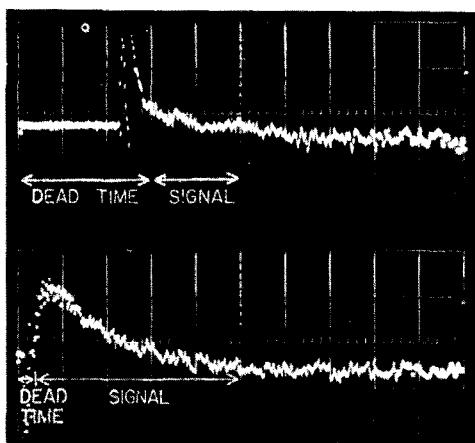


FIG. 8. Performance of the probe damper at 4.3 MHz at the signal processing level. The horizontal scale is in each case $50 \mu\text{sec}/\text{div}$. (a) Without probe damper in use; (b) with probe damper in use.

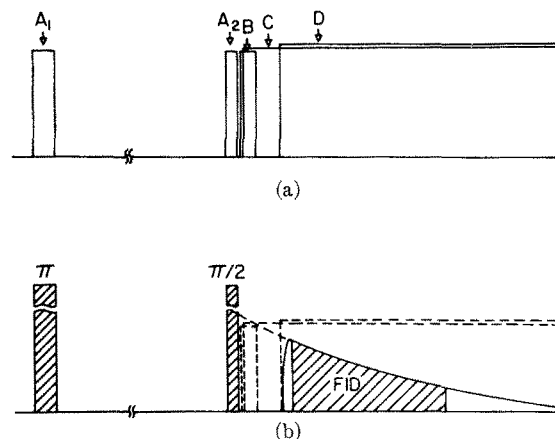


FIG. 9(a). The sequences of pulses used to activate the coupling scheme and probe damper. Pulses A_1 and A_2 gate the transmitter for π and $\pi/2$ pulses, B gates the probe damper, C the coupling scheme, and D the receiver switches and the signal averaging system; (b) illustration of the receiver response to the sequence of gating pulses.

effects in the resonant circuit. A minimum dead time of just less than $20 \mu\text{sec}$ is reasonable.

At 15.78 MHz, a damping factor of ~ 3 is achieved, at 49 MHz no significant reduction of the dead time is expected.

IV. PULSE REQUIREMENTS

The sequence of gating pulses used to activate the coupling scheme and the probe damper is illustrated in Fig. 9(a). Pulses A_1 and A_2 gate the transmitter to produce π and $\pi/2$ pulses, respectively. Pulses B and C gate the probe damper and coupling scheme, respectively, and pulse D gates the receiver internally and triggers the signal averaging system. Pulses B, C, and D are triggered from the falling edge of pulse A_2 ; they do not have to be adjusted during the course of an experiment. Figure 9(b) indicates how the receiver observes the nuclear free induction decay signal.

The driver output to the PIN diode switch in the receiver gate has a long risetime ($5\text{--}10 \mu\text{sec}$). This protects the receiver until the spurious effects following the $\pi/2$ pulse have disappeared and at the same time avoids the production of gating transients in the tuned circuit of the pre-amplifier. The driver output to the transmitter gate should have a risetime of $\sim 1 \mu\text{sec}$. The driver output to the probe damper PIN diode should have a fast risetime but a slow falltime ($\sim 5 \mu\text{sec}$) to avoid gating transients in the resonant circuit of the probe.

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